

SIMATS ENGINEERING P.L.OAVIKSHA

(192572001)

ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence COURSE CODE: CSA1715

COURSE OUTCOME COVERED

CO3: *Analyze knowledge representation and reasoning using propositional logic, FOL, inference, resolution, chaining methods, knowledge representation to perform logical reasoning for drawing valid conclusions and decision making. (BL4)*

Assessment Tool Weightage: 20%

Duration :1 Hour

QUESTIONS: 4

Total Marks:30

Annexure A

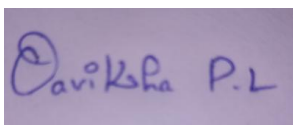
Resolution Algorithm Implementation

Code of Conduct:

I _P.L.OAVIKSHA(192572001) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



Annexure A

Resolution Algorithm Implementation

Question 1 – Rain and Wet Ground

Knowledge Base

1. If it rains, then the ground becomes wet.
2. It is raining.

Goal

Prove that **the ground is wet** using the Resolution Algorithm.

Tasks

1. Represent the statements in propositional logic.
2. Convert all statements into CNF.

Negate the goal.

1. Apply the Resolution Algorithm step by step.
2. Show the Empty Clause ($\square$) if the goal is proved.

Question 2 – Student Assignment Submission

Knowledge Base

1. If a student submits the assignment, then the student receives internal marks
2. Rahul submitted the assignment.

Goal

Prove that **Rahul receives internal marks** using the Resolution Algorithm.

Tasks

1. Write the propositional logic expressions.
2. Convert the premises into CNF.

Negate the goal.

1. Apply the Resolution Algorithm.
2. Conclude whether the goal is proved.

Question 3 – Library Membership

Knowledge Base

1. If a person is a library member, then the person can borrow books.
2. Priya is a library member.

Goal

1. Prove that **Priya can borrow books** using the Resolution Algorithm.

Tasks

1. Represent the knowledge base using propositional logic.
2. Convert each statement into CNF.

Negate the goal.

1. Perform resolution until no new clauses can be generated.
2. State the final conclusion.

Question 4 – Placement Eligibility

Knowledge Base

1. If a student clears the aptitude test, then the student is eligible for placement.
2. Arun cleared the aptitude test.

Goal

1. Prove that **Arun is eligible for placement** using the Resolution Algorithm.

Tasks

1. Convert the statements into propositional logic.

2. Write the CNF clauses.
3. Negate the goal.
4. Apply the Resolution Algorithm step by step.
5. Draw the resolution tree and conclude.

Question 5 – Access Control System

Knowledge Base

1. If a user enters the correct password, then the user is authenticated.
2. If a user is authenticated, then the user is granted access.
3. The user entered the correct password.

Goal

Prove that **the user is granted access** using the Resolution Algorithm.

Tasks

1. Represent the knowledge base in propositional logic.
2. Convert all implications into CNF.
3. Negate the goal.
4. Apply the Resolution Algorithm to derive the Empty Clause ($\square$).
5. Explain each resolution step and write the final conclusion

RUBRICS FOR CALCULATION

Numerical Problems					
Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation

Question 1 – Rain and Wet Ground

Step 1: Representing the statements in propositional logic

We start by assigning simple propositional symbols to the two atomic statements in the knowledge base. Let R stand for "It rains" and W stand for "The ground is wet." The first statement, "If it rains, then the ground becomes wet," is a conditional relationship, so it translates directly into the implication $R \rightarrow W$. The second statement, "It is raining," is simply asserted as a fact, so it becomes R on its own.

Step 2: Converting into CNF (Conjunctive Normal Form)

To use the Resolution Algorithm, every sentence in the knowledge base must be expressed as a disjunction of literals (a clause). The implication $R \rightarrow W$ is not yet in this form, so we apply the standard logical equivalence for eliminating implications: $A \rightarrow B$ is logically equivalent to $\neg A \vee B$. Applying this rule to $R \rightarrow W$ gives us $\neg R \vee W$. The second statement, R , is already a single literal, so it is already a valid clause on its own. This gives us our two CNF clauses:

- C1: $\neg R \vee W$
- C2: R

Step 3: Negating the goal

The goal we want to prove is W ("the ground is wet"). The Resolution Algorithm works by proof through contradiction — instead of proving the goal directly, we assume its opposite is true and show that this assumption leads to a logical contradiction (the empty clause). So we negate the goal, giving us the additional clause C3: $\neg W$, and add it to our knowledge base.

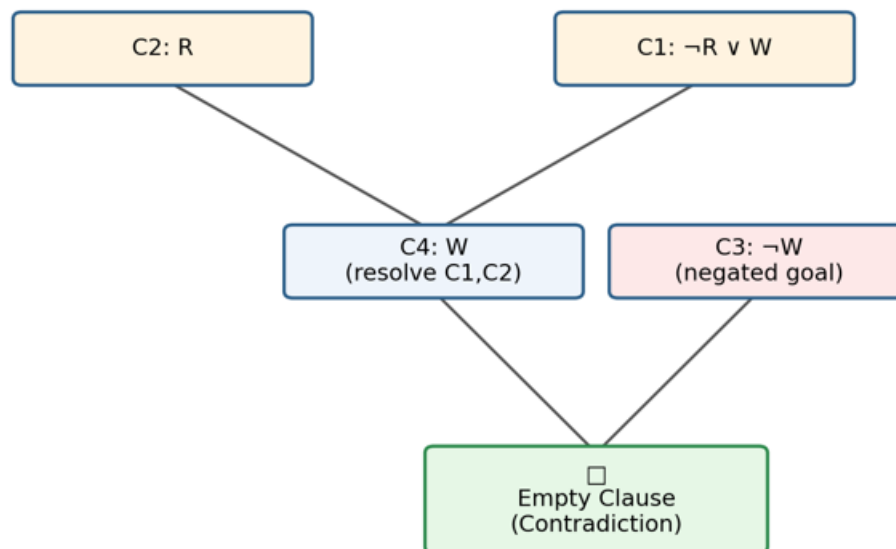
Step 4: Applying the Resolution Algorithm step by step

Resolution works by taking two clauses that contain complementary literals (one clause has a literal, and the other has its negation), cancelling those two literals, and combining whatever remains into a new clause.

- First, look at C2 (R) and C1 ($\neg R \vee W$). These two clauses share the complementary pair R and $\neg R$. Cancelling them out leaves just W . This becomes a new derived clause, C4: W .

- Next, look at C4 (W) and C3 ($\neg W$). These two clauses are direct complements of each other — one asserts W is true, the other asserts W is false. Resolving them cancels out completely, leaving nothing behind. This "nothing" is the empty clause, written as $\square$.

Step 5: Diagram of the resolution process



Conclusion: Since resolving our clauses (including the negated goal) eventually produces the empty clause, this means our assumption that $\neg W$ was true led to a contradiction. Therefore, the original goal W must be true — the ground is wet. The goal is formally PROVED using resolution.

Question 2 – Student Assignment Submission

Step 1: Propositional logic representation

Let S represent "A student submits the assignment" and M represent "The student receives internal marks." The first knowledge base statement, being a conditional, becomes $S \rightarrow M$. The second statement, which asserts a specific fact about Rahul, becomes simply S (interpreting S here as "Rahul submitted the assignment" for this specific instance).

Step 2: Converting the premises into CNF

Using the same implication-elimination rule ($A \rightarrow B \equiv \neg A \vee B$), the first premise $S \rightarrow M$ becomes $\neg S \vee M$. The second premise, S , is already a single literal and needs no further conversion. Our CNF clause set is:

- C1: $\neg S \vee M$
- C2: S

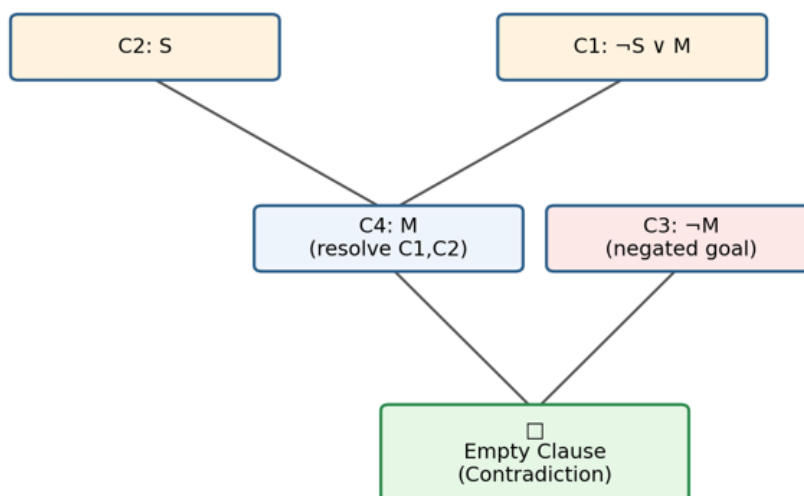
Step 3: Negating the goal

The goal is to prove M ("Rahul receives internal marks"). We negate this to get C3: $\neg M$, which we temporarily assume to be true in order to search for a contradiction.

Step 4: Applying the Resolution Algorithm

- Resolve C2 (S) with C1 ($\neg S \vee M$): the S and $\neg S$ cancel each other, leaving M as a new clause, C4.
- Resolve C4 (M) with C3 ($\neg M$): these are direct complements, and resolving them produces the empty clause $\square$.

Diagram



Step 5: Conclude whether the goal is proved

Because the resolution process terminated in the empty clause, the assumption $\neg M$ leads to a logical contradiction with the rest of the knowledge base. This means the goal M cannot be false given the premises, so it must be true. The goal — Rahul receives internal marks — is formally PROVED.

Question 3 – Library Membership

Step 1: Representing the knowledge base using propositional logic

Let L represent "A person is a library member" and B represent "The person can borrow books." The first statement is conditional and becomes $L \rightarrow B$. The second statement asserts a fact about Priya specifically, so it becomes L (read here as "Priya is a library member").

Step 2: Converting each statement into CNF

Applying the implication-elimination equivalence to $L \rightarrow B$ gives $\neg L \vee B$. The fact L remains as is, since a single literal already satisfies the CNF requirement of being a disjunction of literals (a disjunction of just one literal is trivially valid). Our clauses are:

- $C1: \neg L \vee B$
- $C2: L$

Step 3: Negating the goal

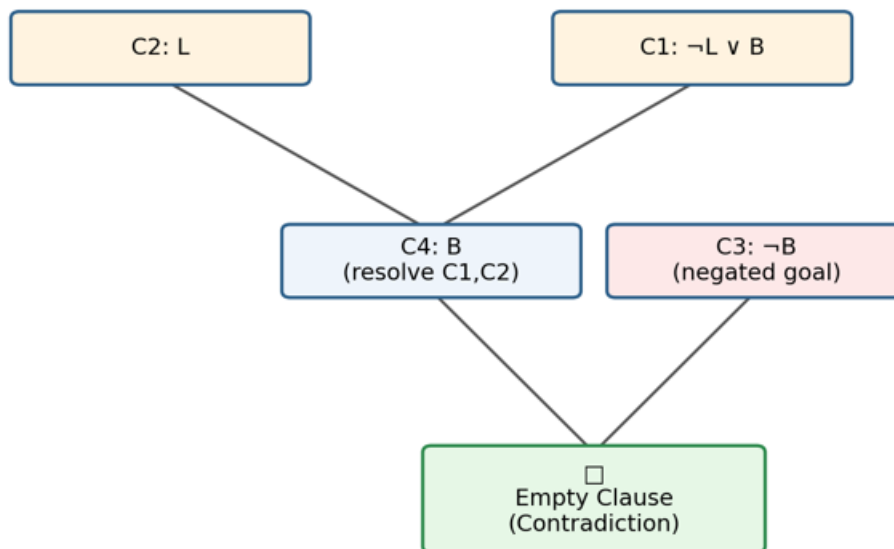
The goal is B ("Priya can borrow books"). Negating this gives $C3: \neg B$, which we add to the clause set to attempt a proof by contradiction.

Step 4: Performing resolution until no new clauses can be generated

- Resolve $C2 (L)$ with $C1 (\neg L \vee B)$: the complementary pair L and $\neg L$ cancels, leaving B as a new clause, $C4$.
- Resolve $C4 (B)$ with $C3 (\neg B)$: these directly contradict one another, and resolving them yields the empty clause $\square$.

At this point, no further clauses can meaningfully be generated because the empty clause represents the termination condition of the algorithm — once a contradiction is found, the search stops.

Diagram



Step 5: State the final conclusion

Since the negation of the goal led to a contradiction with the knowledge base, the original goal must be true. Therefore, it is formally PROVED that Priya can borrow books.

Question 4 – Placement Eligibility

Step 1: Converting the statements into propositional logic

Let A represent "A student clears the aptitude test" and E represent "The student is eligible for placement." The first statement is a conditional relationship: $A \rightarrow E$. The second is a factual assertion about Arun specifically: A (interpreted as "Arun cleared the aptitude test").

Step 2: Writing the CNF clauses

Applying the implication rule ($A \rightarrow B \equiv \neg A \vee B$) to $A \rightarrow E$ gives $\neg A \vee E$. The second statement A is already a single-literal clause. So:

- C1: $\neg A \vee E$
- C2: A

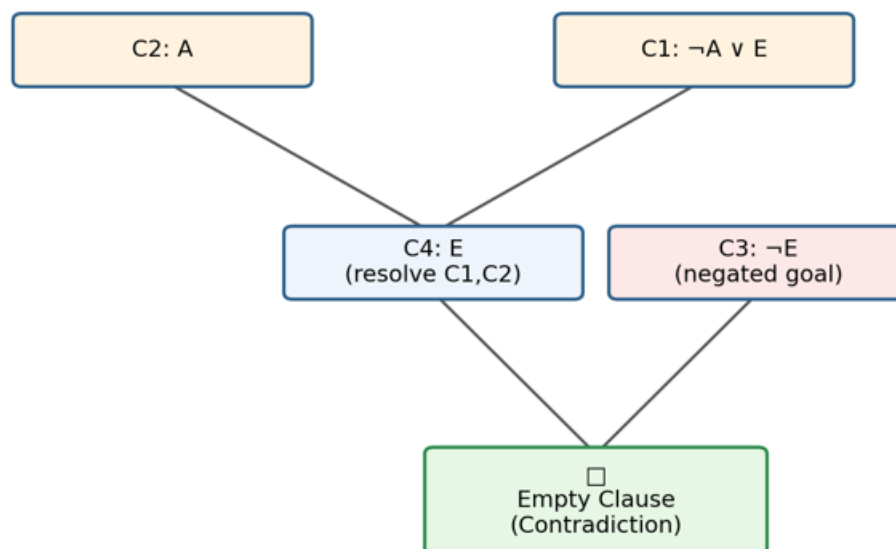
Step 3: Negating the goal

The goal is E ("Arun is eligible for placement"). We negate it to obtain C3: $\neg E$, which is added to the knowledge base as our contradiction-seeking assumption.

Step 4: Applying the Resolution Algorithm step by step

- Resolve C2 (A) with C1 ($\neg A \vee E$): the A and $\neg A$ literals cancel out, leaving E as a newly derived clause, C4.
- Resolve C4 (E) with C3 ($\neg E$): E and $\neg E$ are exact complements, and resolving them leaves nothing producing the empty clause $\square$.

Step 5: Drawing the resolution tree and concluding



Each branch of the tree represents one resolution step: the first branch combines the fact A with the implication clause to derive E; the second branch combines that derived E with the negated goal $\neg E$ to reach the empty clause. Because the tree terminates in $\square$, the assumption that Arun is not eligible for placement is contradicted by the knowledge base. Therefore, the goal is formally PROVED Arun is eligible for placement.

Question 5 – Access Control System

Step 1: Representing the knowledge base in propositional logic

This knowledge base has three statements, forming a two-step implication chain. Let P represent "The user enters the correct password," $Auth$ represent "The user is authenticated," and Acc represent "The user is granted access." The first statement becomes $P \rightarrow Auth$. The second becomes $Auth \rightarrow Acc$. The third is a direct fact: P .

Step 2: Converting all implications into CNF

Each implication is converted using the equivalence $A \rightarrow B \equiv \neg A \vee B$:

- $P \rightarrow Auth$ becomes $\neg P \vee Auth$
- $Auth \rightarrow Acc$ becomes $\neg Auth \vee Acc$
- P remains as a single-literal clause

This gives us:

- $C1: \neg P \vee Auth$
- $C2: \neg Auth \vee Acc$
- $C3: P$

Step 3: Negating the goal

The goal is Acc ("the user is granted access"). We negate it to get $C4: \neg Acc$, which we add to the clause set as the assumption we intend to disprove.

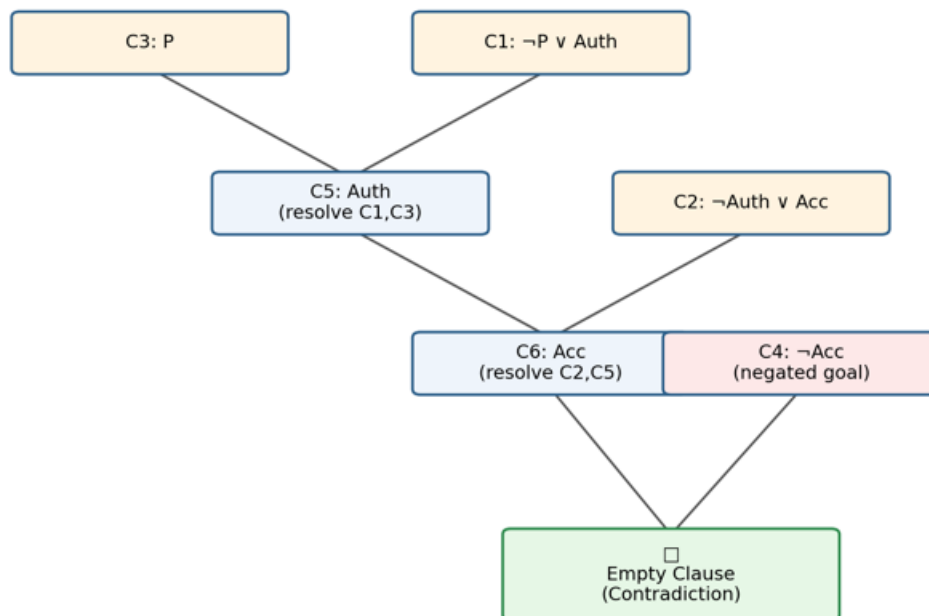
Step 4: Applying the Resolution Algorithm to derive the Empty Clause

Since this problem involves a two-step chain, resolution must be applied twice before reaching the negated goal:

- **Step A:** Resolve $C3 (P)$ with $C1 (\neg P \vee Auth)$. The complementary pair P and $\neg P$ cancels out, leaving $Auth$ as a new derived clause, $C5$. This step essentially says: "Since the user entered the correct password, and correct-password implies authenticated, the user must be authenticated."

- **Step B:** Resolve C5 (Auth) with C2 ($\neg \text{Auth} \vee \text{Acc}$). The complementary pair Auth and $\neg \text{Auth}$ cancels out, leaving Acc as a new derived clause, C6. This step says: "Since the user is authenticated, and authenticated implies granted access, the user must be granted access."
- **Step C:** Resolve C6 (Acc) with C4 ($\neg \text{Acc}$). These are direct complements of each other, so resolving them cancels out entirely, producing the empty clause $\square$.

Step 5: Diagram and explanation of each resolution step



Reading the tree from top to bottom: the first resolution step combines the known fact "password entered" with the implication "password implies authentication" to conclude that the user is authenticated. The second resolution step takes that conclusion and combines it with the implication "authentication implies access" to conclude that the user is granted access. The final resolution step takes this derived conclusion and combines it with the negated goal, producing a direct contradiction (the empty clause).

Final conclusion: Because assuming the negation of the goal ($\neg \text{Acc}$) leads step by step to a logical contradiction with the knowledge base, the original goal cannot be false. Therefore, it is formally PROVED that the user is granted access.